Ertapenem Kabi Powder for Solution for Infusion or Injection 1g/vial

DESCRIPTION:
White to yellowish powder.

Reconstituted/Diluted solution:
The reconstituted solutions should be inspected visually for particulate matter and discoloration prior to administration, whenever the container permits. Solutions of Ertapenem Kabi range from colourless to pale yellow. Variations of colour within this range do not affect potency.

Administration route	Reconstitution product	Dilution product
IV	Sterile Water for Injection	0.9% Sodium Chloride for Injection
	0.9% Sodium Chloride for Injection	0.9% Sodium Chloride for Injection
	Bacteriostatic Water for Injection	0.9% Sodium Chloride for Injection
IM	1.0 % Lidocaine HCl for Injection (without epinephrine)	--
	2.0 % Lidocaine HCl for Injection (without epinephrine)	--

DOSAGE FORM:

Powder for solution for infusion or injection.

COMPOSITION:
Each vial contains Ertapenem (as sodium) 1gm

PHARMACOLOGY:
Ertapenem has in vitro activity against a wide range of gram positive and gram-negative aerobic and anaerobic bacteria. The bactericidal activity of ertapenem results from the inhibition of cell wall synthesis and is mediated through ertapenem binding to penicillin binding proteins (PBPs). In *Escherichia coli*, it has strong affinity toward PBPs 1a, 1b, 2, 3, 4 and 5 with preference for PBPs 2 and 3. Ertapenem has significant stability to hydrolysis by most classes of beta-lactamases, including penicillinases, and cephalosporinases and extended spectrum beta-lactamases, but not metallo-beta-lactamases.

PHARMACOKINETICS:

Absorption:
Ertapenem, reconstituted with 1% lidocaine HCl injection, USP (in saline without epinephrine), is well absorbed following IM administration at the recommended dose of 1 g. The mean bioavailability is approximately 92%. Following 1 g daily IM administration, mean peak plasma concentrations (Cmax) are reached in approximately 2 hours (Tmax).

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Distribution:
Ertapenem is highly bound to human plasma proteins. In healthy young adults, the protein binding of Ertapenem decreases as plasma concentrations increase, from approximately 95% bound at an approximate plasma concentration of <100 micrograms (mcg)/mL to approximately 85% bound at an approximate plasma concentration of 300 mcg/mL.

Area under the plasma concentration curve (AUC) of Ertapenem in adults increases nearly dose-proportionally over the 0.5 to 2 g dose range.
There is no accumulation of Ertapenem in adults following multiple IV doses ranging from 0.5 to 2 g daily or IM doses of 1 g daily.
The volume of distribution (Vdss) of Ertapenem in adults is approximately 8 liters (0.11 liter/kg) and approximately 0.2 liter/kg in pediatric patients 3 months to 12 years of age and approximately 0.16 liter/kg in pediatric patients 13 to 17 years of age.
Ertapenem penetrates into suction-induced skin blisters.
Ertapenem does not inhibit P-glycoprotein-mediated transport of digoxin or vinblastine and that Ertapenem is not a substrate for P-glycoprotein-mediated transport

Metabolism:
The major metabolite of Ertapenem is the ring-opened derivative formed by hydrolysis of the beta-lactam ring.
Human liver microsomes indicate that Ertapenem does not inhibit metabolism mediated by any of the six major cytochrome p450 (CYP) isoforms: 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4.

Elimination:
Ertapenem is eliminated primarily by the kidneys. The mean plasma half-life in healthy young adults and patients 13 to 17 years of age is approximately 4 hours and approximately 2.5 hours in pediatric patients 3 months to 12 years of age.
Following administration of a 1 g IV dose of ertapenem to healthy young adults, approximately 80% is recovered in urine and 10% in feces. Of the 80% recovered in urine, approximately 38% is excreted as unchanged drug and approximately 37% as the ring-opened metabolite.
In healthy young adults given a 1 g IV dose, average concentrations of Ertapenem in urine exceed 984 mcg/mL during the period 0 to 2 hours postdose and exceed 52 mcg/mL during the period 12 to 24 hours postdose.

INDICATIONS:

Treatment

Ertapenem Kabi is indicated for the treatment of patients with moderate to severe infections caused by susceptible strains of microorganisms, as well as initial empiric therapy prior to the identification of causative organisms in the infections listed below:

- Complicated Intra-Abdominal Infections
- Complicated Skin and Skin Structure Infections including diabetic lower extremity and diabetic foot infections
- Community Acquired Pneumonia
- Complicated Urinary Tract Infections including pyelonephritis
- Acute Pelvic Infections including postpartum endomyometritis, septic abortion and post surgical gynecologic infections

Ertapenem Kabi is indicated for the treatment of pediatric patients with moderate to severe infections caused by susceptible strains of microorganisms, as well as initial empiric therapy prior to identification of causative organisms in the indications listed below:-

- Complicated intra-abdominal infections

- Complicated skin and skin structure infections
- Community acquired pneumonia
- Acute pelvic infections including postpartum endomyometritis, septic abortion and surgical gynaecologic infections

Prevention

Ertapenem Kabi is indicated in adults for the prophylaxis of surgical site infection following elective colorectal surgery.

RECOMMENDED DOSE:

The usual dose of Ertapenem Kabi in patients 13 years of age and older is 1 gram (g) given once a day. The usual dose of Ertapenem Kabi in patients 3 months to 12 years of age is 15 mg/kg twice daily (not to exceed 1 g/day).

Ertapenem Kabi may be administered by intravenous (IV) infusion or intramuscular (IM) injection. When administered intravenously, ertapenem should be infused over a period of 30 minutes.
Intramuscular administration of Ertapenem Kabi may be used as an alternative to intravenous administration in the treatment of those infections for which intramuscular therapy is appropriate.
The usual duration of therapy with Ertapenem Kabi is 3 to 14 days but varies by the type of infection and causative pathogen(s). When clinically indicated, a switch to an appropriate oral antimicrobial may be implemented if clinical improvement has been observed.
In controlled clinical studies, patients were treated from 3 to 14 days. Total treatment duration was determined by the treating physician based on site and severity of the infection, and on the patient's clinical response. In some studies, treatment was converted to oral therapy at the discretion of the treating physician after clinical improvement had been demonstrated.

Prophylaxis of surgical site infection following elective colorectal surgery:

To prevent surgical site infections following elective colorectal surgery in adults, the recommended dosage is 1 g IV administered as a single intravenous dose given 1 hour prior to the surgical incision.

Patients with Renal Insufficiency:

Ertapenem Kabi may be used for the treatment of infections in adult patients with renal insufficiency. In patients whose creatinine clearance is > 30 mL/min/1.73 m^2, no dosage adjustment is necessary. Adult patients with advanced renal insufficiency (creatinine clearance $\leq$ 30 mL/min/1.73 m^2), including those on hemodialysis, should receive 500 mg daily. There are no data in pediatric patients with renal insufficiency.

Patients on Hemodialysis:

Following a single 1 g IV dose of Ertapenem Kabi given immediately prior to a hemodialysis session, approximately 30% of the dose was recovered in the dialysate. When adult patients on hemodialysis are given the recommended daily dose of 500 mg of Ertapenem Kabi within 6 hours prior to hemodialysis, a supplementary dose of 150 mg is recommended following the hemodialysis session. If Ertapenem Kabi is given at least 6 hours prior to hemodialysis, no supplementary dose is needed. There are no data in patients undergoing peritoneal dialysis or hemofiltration. There are no data in pediatric patients on hemodialysis.

When only the serum creatinine is available, the following formula may be used to estimate creatinine clearance. The serum creatinine should represent a steady state of renal function.

$$\text{Males} : \frac{(\text{weight in kg}) \times (140\text{-age in years})}{(72) \times \text{serum creatinine (mg/100 mL)}}$$

$$\text{Females} : (0.85) \times (\text{value calculated for males})$$

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No dosage adjustment is recommended in patients with impaired hepatic function.
The recommended dose of ertapenem can be administered without regard to age (13 years and older) or gender.

CONTRAINDICATIONS:

Ertapenem is contraindicated in patients with known hypersensitivity to any component of this product or to other drugs in the same class or in patients who have demonstrated anaphylactic reactions to beta-lactams.
Due to the use of lidocaine HCl as a diluent, ertapenem administered intramuscularly is contraindicated in patients with a known hypersensitivity to local anesthetics of the amide type and in patients with severe shock or heart block. (Refer to the prescribing information for lidocaine HCl.)

ADVERSE EFFECTS:

Adult Patients

The most common drug-related adverse experiences reported during parenteral therapy in patients treated with ertapenem were diarrhea ,infused vein complication ,nausea and headache.
The following drug-related adverse experiences were reported during parenteral therapy in adult patients treated with ertapenem:

Common	Nervous system disorders	Headache
	Vascular disorders	Infused vein complication, phlebitis/thrombophlebitis
	Gastrointestinal disorders	Diarrhea, nausea, vomiting
Uncommon	Nervous system disorders	Dizziness, somnolence, insomnia, seizure, confusion
	Cardiac and vascular disorders	Extravasation, hypotension
	Respiratory, thoracic and mediastinal disorders	Dyspnea
	Gastrointestinal disorders	Oral candidiasis, constipation, acid regurgitation, *C. difficile*-associated diarrhea, dry mouth, dyspepsia, anorexia
	Skin and subcutaneous tissue disorders	Erythema, pruritus
	General disorders and administration site conditions	Abdominal pain, taste perversion, asthenia/fatigue, candidiasis, edema/swelling, fever, pain, chest pain
	Reproductive system and breast disorders	Vaginal pruritus

Seizure was reported during parenteral therapy in patients treated with ertapenem, piperacillin/tazobactam and ceftriaxone.

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Pediatric Patients

The overall safety profile in pediatric patients is comparable to that in adult patients. The most common drug-related clinical adverse experiences reported during parenteral therapy were diarrhea infusion site pain and infusion site erythema

The following drug-related adverse experiences were reported during parenteral therapy in pediatric patients treated with ertapenem:

Common	Gastrointestinal disorders	Diarrhea, vomiting
	General disorders and administration site conditions	Infusion site erythema, infusion site pain, infusion site phlebitis, infusion site swelling
	Skin and subcutaneous tissue disorders	Rash

Additional drug-related adverse experiences that were reported during parenteral therapy in patients treated with ertapenem in include: infusion site induration, infusion site pruritus, infusion site warmth and phlebitis.

DRUG INTERACTIONS:

When ertapenem is administered with probenecid, probenecid competes for active tubular secretion and thus inhibits the renal excretion of ertapenem. This leads to small but statistically significant increases in the elimination half-life (19%) and in the extent of systemic exposure (25%). No dosage adjustment is necessary when ertapenem is given with probenecid. Because of the small effect on half-life, the co-administration with probenecid to extend the half-life of ertapenem is not recommended.

Ertapenem does not inhibit P-glycoprotein-mediated transport of digoxin or vinblastine and that ertapenem is not a substrate for P-glycoprotein-mediated transport. In vitro studies in human liver microsomes indicate ertapenem does not inhibit metabolism mediated by any of the six major cytochrome p450 (CYP) isoforms: 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4. Drug interactions caused by inhibition of P-glycoprotein-mediated drug clearance or CYP-mediated drug clearance are unlikely. Other than with probenecid, no specific clinical drug interaction studies have been conducted.

Case reports in the literature have shown that co-administration of carbapenems, including ertapenem, to patients receiving valproic acid or divalproex sodium results in a reduction of valproic acid concentrations. The valproic acid concentrations may drop below the therapeutic range as a result of this interaction, therefore increasing the risk of breakthrough seizures. Although the mechanism of this interaction is unknown, data from in vitro and animal studies suggest that carbapenems may inhibit the hydrolysis of valproic acid's glucuronide metabolite (VPA-g) back to valproic acid, thus decreasing the serum concentrations of valproic acid.

PRECAUTIONS AND WARNINGS:

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. These reactions are more likely to occur in individuals with a history of sensitivity to multiple allergens. There have been reports of individuals with a history of penicillin hypersensitivity who have experienced severe hypersensitivity reactions when treated with another beta-lactam. Before initiating therapy with ertapenem, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems, other beta-lactams and other allergens. If an allergic reaction to ertapenem occurs, discontinue the drug immediately and institute appropriate alternative therapy. Serious anaphylactic reactions require immediate emergency treatment.

Seizures and other central nervous system (CNS) adverse experiences have been reported during treatment with ertapenem. During clinical investigations in adult patients treated with ertapenem (1 g once a day), seizures, irrespective of drug relationship, occurred in 0.5% of patients during study therapy plus 14-day follow-up period. These experiences have occurred most commonly in patients with CNS disorders (e.g., brain lesions or history of seizures) and/or compromised renal function. Close adherence to the recommended dosage regimen is urged, especially in patients with known factors that predispose to convulsive activity. Anticonvulsant therapy should be continued in patients with known seizure disorders. If focal tremors, myoclonus, or seizures occur, patients should be evaluated neurologically and the dosage of ertapenem re-examined to determine whether it should be decreased or discontinued.

Case reports in the literature have shown that co-administration of carbapenems, including ertapenem, to patients receiving valproic acid or divalproex sodium results in a reduction in valproic acid concentrations. The valproic acid concentrations may drop below the therapeutic range as a result of this interaction, therefore increasing the risk of breakthrough seizures. Increasing the dose of valproic acid or divalproex sodium may not be sufficient to overcome this interaction. The concomitant use of ertapenem and valproic acid/divalproex sodium is generally not recommended. Anti-bacterials other than carbapenems should be considered to treat infections in patients whose seizures are well controlled on valproic acid or divalproex sodium. If administration of ertapenem is necessary, supplemental anti-convulsant therapy should be considered.

As with other antibiotics, prolonged use of ertapenem may result in overgrowth of non-susceptible organisms. Repeated evaluation of the patient's condition is essential. If superinfection occurs during therapy, appropriate measures should be taken.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including ertapenem, and may range in severity from mild to life-threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of antibacterial agents. Studies indicate that a toxin produced by *Clostridium difficile* is a primary cause of “antibiotic-associated colitis”.

Caution should be taken when administering ertapenem intramuscularly, to avoid inadvertent injection into a blood vessel.

Lidocaine HCl is the diluent for intramuscular administration of ertapenem. Refer to the prescribing information for lidocaine HCl.

Ertapenem Kabi is not recommended in infants under 3 months of age as no data are available.

The efficacy and safety of ertapenem in the elderly (> 65 years) was comparable to that seen in younger patients (< 65 years).

FERTILITY, PREGNANCY AND LACTATION:

Pregnancy

There are no adequate and well-controlled studies in pregnant women. Ertapenem should be used during pregnancy only if the potential benefit justifies the potential risk to the mother and fetus.

Lactation

Ertapenem is excreted in human milk. Caution should be exercised when ertapenem is administered to a nursing woman.

OVERDOSE:
No specific information is available on the treatment of overdosage with ertapenem. Intentional overdosing of ertapenem is unlikely. Intravenous administration of ertapenem at a 3 g daily dose for 8 days to healthy adult volunteers did not result in significant toxicity. In adults, inadvertent administration of up to 3 g in a day did not result in clinically important adverse experiences. In pediatric, a single IV dose of 40 mg/kg up to a maximum of 2 g did not result in toxicity.
In the event of an overdose, ertapenem should be discontinued and general supportive treatment given until renal elimination takes place.
Ertapenem can be removed by hemodialysis; however, no information is available on the use of hemodialysis to treat overdosage.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
No studies on the effects on the ability to drive and use machines have been performed. Ertapenem may influence patients' ability to drive and use machines. Patients should be informed that dizziness and somnolence have been reported with Ertapenem.

STORAGE:
Unopened Vial: Do not store above 25°C. For Single Use.

INSTRUCTIONS FOR USE
For single use only.

Patients 13 years of age and older

<u>Preparation for intravenous administration:</u>
DO NOT MIX OR CO-INFUSE ERTAPENEM KABI WITH OTHER MEDICATIONS.
DO NOT USE DILUENTS CONTAINING DEXTROSE (α-D-GLUCOSE)

ErtapenemKabi must be reconstituted and then diluted prior to administration.

<u>Reconstitution</u>
Reconstitute the contents of a 1 g vial of Ertapenem Kabi with 10 ml of the following:Water for injection, 0.9% Sodium Chloride Injection or Bacteriostatic Water For Injection. Shake well to dissolve it.

<u>Dilution</u>
For a 50 ml bag of diluent: For a 1 g dose, immediately transfer contents of the reconstituted vial to a 50 ml bag of sodium chloride 9 mg/ml (0.9 %) solution; or

For a 50 ml vial of diluent: For a 1 g dose, immediately withdraw 10 ml from a 50 ml vial of sodium chloride 9 mg/ml (0.9 %) solution and discard. Immediately transfer the contents of the reconstituted 1 g vial of Ertapenem Kabi to the 50 ml vial of sodium chloride 9 mg/ml (0.9 %) solution.

<u>Infusion</u>
Infuse over a period of 30 minutes.

<u>Preparation for intramuscular administration:</u>
Ertapenem Kabi must be reconstituted prior to administration.

Reconstitute the contents of a 1 g vial of Ertapenem Kabi with 3.2 mL of 1.0% or 2.0% lidocaine HCl injection (without epinephrine). Shake vial thoroughly to form solution.

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Immediately withdraw the contents of the vial and administer by deep intramuscular injection into a large muscle mass (such as the gluteal muscles or lateral part of the thigh).

The reconstituted IM solution should be used within 1 hour after preparation.
Note: The reconstituted solution should not be administered intravenously.

Pediatric (3 months to 12 years of age)

Preparation for intravenous administration:
DO NOT MIX OR CO-INFUSE ERTAPENEM KABI WITH OTHER MEDICATIONS.
DO NOT USE DILUENTS CONTAINING DEXTROSE (α-D-GLUCOSE)
Ertapenem Kabi must be reconstituted and then diluted prior to administration.

Reconstitution
Reconstitute the contents of a 1 g vial of Ertapenem Kabi with 10 ml of the following:Water for injection, 0.9% Sodium Chloride Injection or Bacteriostatic Water For Injection. Shake well to dissolve it.

Dilution
For a bag of diluent: Immediately transfer a volume equal to 15 mg/kg of body weight (not to exceed 1 g/day) to a bag of sodium chloride 9 mg/ml (0.9 %) solution for a final concentration of 20 mg/ml or less; or

For a vial of diluent: Immediately transfer a volume equal to 15 mg/kg of body weight (not to exceed 1 g/day) to a vial of sodium chloride 9 mg/ml (0.9 %) solution for a final concentration of 20 mg/ml or less.

Infusion
Infuse over a period of 30 minutes.

Preparation for intramuscular administration:
Ertapenem Kabi must be reconstituted prior to administration.

Reconstitute the contents of a 1 g vial of Ertapenem Kabi with 3.2 mL of 1.0% or 2.0% lidocaine HCl injection (without epinephrine). Shake vial thoroughly to form solution.

Immediately withdraw a volume equal to 15 mg/kg of body weight (not to exceed 1 g/day) and administer by deep intramuscular injection into a large muscle mass (such as the gluteal muscles or lateral part of the thigh).

The reconstituted IM solution should be used within 1 hour after preparation. **Note: The reconstituted solution should not be administered intravenously.**

The parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever the container permits. Solutions of Ertapenem Kabi range from colourless to pale yellow. Variations of colour within this range do not affect potency.

Any unused product or waste material should be disposed of in accordance with local requirements.

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INCOMPATIBILITIES:
Do not use solvents or infusion fluids containing dextrose for reconstitution or administration of ertapenem.
In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products except those mentioned in section Pharmaceutical precautions.

SHELF LIFE:

Unopened vial
24 months. For single use only.

After reconstitution
The reconstituted IM solution should be used within 1 hour after preparation.
The reconstituted IV solutions should be diluted immediately after preparation.

After dilution
Chemical and physical in-use stability for diluted solutions has been demonstrated for 6 hours at 25°C or for 24 hours at 2 to 8°C (in a refrigerator). Solutions should be used within 4 hours of their removal from the refrigerator. Do not freeze solutions of ertapenem. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2 to 8°C, unless reconstitution/dilution (etc.) has taken place in controlled and validated aseptic conditions.

HANDLING AND DISPOSAL:

Ertapenem is for single use only. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

PRESENTATION:
20 ml colorless clear Type I glass vials with a chlorobutyl stopper and an aluminium flip of overseal. Supplied in packs of 10 vials.

Product Registration Holder/Importer:
Fresenius Kabi Malaysia Sdn Bhd
3-1 & 3-2, Axis Technology Centre,
Lot 13, Jalan 51A/225,
46100 Petaling Jaya,
Selangor, Malaysia

MANUFACTURER INFORMATION:

Manufactured by:
ACS Dobfar S.p.a,
Nucleo Industriale S.Atto (loc. S. Nicolo` A Tordino)
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